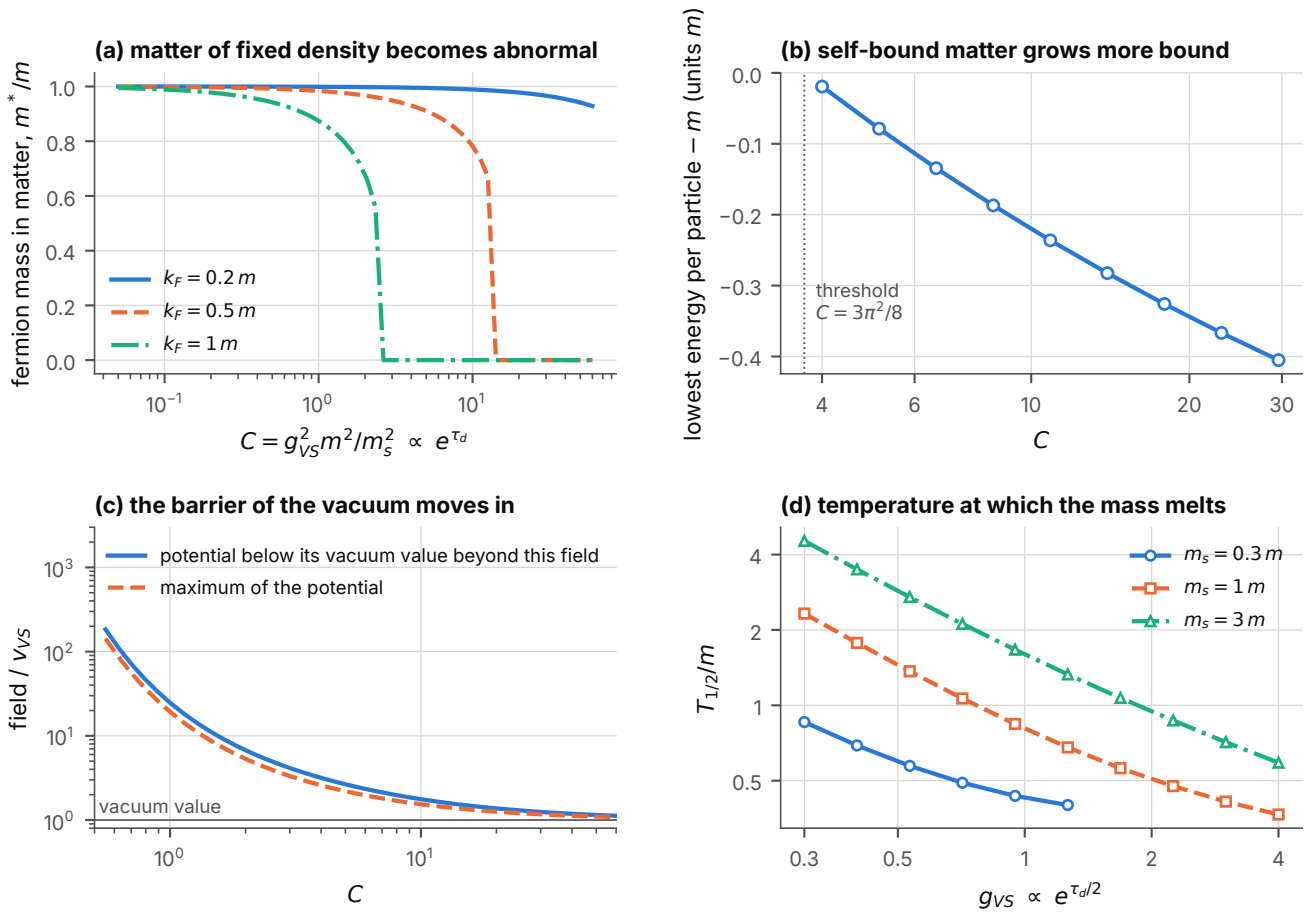
(a) matter of fixed density becomes abnormal
fermion mass in matter, m^*/m
1.0
0.8
0.6
0.4
0.2
0.0
$k_F = 0.2\,m$
$k_F = 0.5\,m$
$k_F = 1\,m$
10^{-1}
10^{0}
10^{1}
$C = g_{VS}^2 m^2/m_s^2 \propto e^{\tau_d}$
(b) self-bound matter grows more bound
lowest energy per particle − m (units m)
0.0
−0.1
−0.2
−0.3
−0.4
threshold
$C = 3\pi^2/8$
4
6
10
20
30
C
(c) the barrier of the vacuum moves in
potential below its vacuum value beyond this field
maximum of the potential
field / v_{VS}
10^3
10^2
10^1
10^0
vacuum value
10^0
10^1
C
(d) temperature at which the mass melts
$T_{1/2}/m$
4
2
1
0.5
$m_s = 0.3\,m$
$m_s = 1\,m$
$m_s = 3\,m$
0.3
0.5
1
2
4
$g_{VS} \propto e^{\tau_d/2}$